KRISHNA PRASHANTH PAULPANDIAN

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SUMMARY

ECE undergraduate specializing in Digital VLSI, with research and project experience spanning approximate computing, processor design, and hardware security. Currently exploring hardware accelerators and RTL-to-GDSII implementation, with an interest in digital IC design and verification roles.

EDUCATION

Vellore Institute of Technology 2023 – Present
B.Tech in Electronics and Communication Engineering Katpadi, India

CGPA: 9.88/10

Prince Srivari Senior Secondary School 2020 – 2023
Class X: 94.6% | *Class XII: 95.6%* Chennai, India

PROJECTS

Hardware Efficient Approximate Sobel Edge Detector for Image Processing Applications

May 2025 – Jun. 2025

- **Summary:** Developed a synthesizable Verilog-based Sobel edge detector for FPGA using an approximate unsigned multiplier to reduce power and area while enabling real-time edge detection.
- **Tools & Technologies:** Verilog HDL, Xilinx Vivado, FPGA, MATLAB

Design of a Novel Error Compensated Mitchell based Logarithmic Multiplier

Mar. 2026 – Apr. 2026

- **Summary:** Proposed an error-compensated Mitchell-based logarithmic multiplier with reduced hardware overhead and improved computational accuracy for Gaussian filtering applications.
- **Publication:** Submitted an extended version of the work to *IEEE Access*; currently under peer review.
- **Tools & Technologies:** Verilog HDL, ModelSim, MATLAB, Synopsys Design Compiler

PUF-Based FPGA Security for Hardware Identification

Apr. 2026 – Jun. 2026

- **Summary:** Designed and synthesized a 32-Ring-Oscillator PUF architecture in Verilog for FPGA-based hardware authentication; developed challenge-response logic and evaluated implementation behavior using Xilinx Vivado.
- **Tools & Technologies:** Verilog HDL, FPGA, Vivado, Hardware Security, PUF, Digital VLSI

RTL-to-GDSII Implementation of an 8-bit Processor

Jul. 2026 – Present

- **Summary:** Designed and implemented an 8-bit processor in Verilog HDL with core modules including the ALU, register file, and control unit. Currently working on ASIC implementation from RTL to GDSII, including synthesis, floorplanning, placement, clock tree synthesis, routing, and signoff.
- **Tools & Technologies:** Verilog HDL, Xilinx Vivado, Genus, Yosys, OpenSTA

RISC-V Based Post-Quantum Cryptography Accelerator

Jul. 2026 – Present

- **Summary:** Developing a RISC-V-based hardware accelerator for post-quantum cryptographic algorithms, focusing on efficient hardware implementation, performance optimization, and secure cryptographic processing.
- **Tools & Technologies:** Verilog HDL, RISC-V

PUBLICATIONS

Area Efficient Approximate Multiplier Based on Novel 4:2 Compressors and Error Correction Logic
Springer Journal, Dec. 2025

- **Summary:** Proposed an area-efficient approximate multiplier using novel 4:2 compressors and error correction logic to reduce hardware overhead while improving computational accuracy.
- **Skills:** Verilog HDL, Approximate Computing, Digital VLSI, Arithmetic Circuit Design, FPGA, Research Writing

Edge AI-Based Real-Time Fault Diagnostics for Industrial Motors Using Lightweight Neural Networks
2nd International Conference on Next Generation Electronics (NEleX-2026)

- **Summary:** Developed an Edge AI-based framework for real-time industrial motor fault diagnosis using lightweight neural networks, enabling efficient and low-latency predictive maintenance.
- **Skills:** Edge AI, Neural Networks, Embedded Systems, Machine Learning, Signal Processing, Technical Writing

INTERNSHIPS

Research Intern May 2025 – Jun. 2025
National Institute of Technology - Tiruchirappalli Tiruchirappalli, India

- Worked under **Dr. M. Bhaskar, HoD - ECE** on Digital VLSI.
- Developed an FPGA-based approximate Sobel Edge Detector and evaluated image quality using PSNR.

SKILLS

HDLs	Verilog HDL, SystemVerilog
Programming	Java, Embedded C, Assembly, TCL, MATLAB
Tools	Quartus Prime, ModelSim, Xilinx Vivado, Keil uVision, Cadence EDA Tools
Core Skills	RTL Design, FPGA based Design Implementation, Computer Architecture, ASIC Design Flow, 8051 Microcontroller

CERTIFICATIONS

System Design through Verilog Sep. 2024
NPTEL - Indian Institute of Technology, Guwahati

Value Added Course on RTL to GDSII Apr. 2026
Vellore Institute of Technology

LEADERSHIP EXPERIENCE

Programme Representative (PR) Oct. 2025 - Present
School of Electronics Engineering, VIT

Executive Member (Student Council) Aug. 2026 - Present
Office of Students Welfare, VIT

HONOURS & AWARDS

Merit Certificates for Academic Excellence

- ***2023–2024:*** *1st Rank* ***2024–2025:*** *2nd Rank* ***2025–2026:*** *1st Rank*